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A FAIRNESS AND BIAS AUDIT SYSTEM PROPOSAL REPORT ON BIASAPERTURE: A DIAGNOSTIC AND EVALUATIVE FRAMEWORK FOR AUDITING DEMOGRAPHIC BIAS IN FACIAL ANALYSIS SYSTEMS

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ABSTRACT

Automated facial analysis systems are widely deployed despite well-documented accuracy disparities across demographic subgroups, and independent, reproducible auditing of these systems remains rare in practice. This report proposes BiasAperture, a diagnostic and evaluative software platform that computes subgroup and intersectional fairness metrics for a third-party facial-analysis model and reports them in a standardised, regulator-legible format. The platform is organised into five cooperating modules covering data ingestion, model interfacing, fairness-metric computation, explainability, and report generation. Its analytical core computes four disparity metrics, demographic parity difference, equalized odds difference, equal opportunity difference, and disparate impact ratio, using AIF360 and Fairlearn as independent, cross-validating backends, with every reported disparity accompanied by a chi-squared significance test and a bootstrap confidence interval rather than a bare threshold. The current explainability implementation uses demographic-dummy surrogate attribution to attribute flagged disparities to input features, distinguishing genuinely demographic effects from unrelated confounds; richer spatial SHAP and ITA analysis remain deferred. Findings are assembled into an exportable, Model-Cards- and Datasheets-structured report in which every metric is traced to its specific basis under Article 10 of the EU AI Act and the corresponding function of the NIST AI Risk Management Framework. The current case study uses FairFace; UTKFace was profiled and cut from the implementation scope, with a FairFace-trained convolutional-network baseline serving as the platform's own case study. BiasAperture is scoped strictly as diagnostic: it identifies and statistically characterises disparities but does not mitigate bias, retrain models, or generate synthetic demographic data.

Keywords: Algorithmic Fairness, Bias Auditing, Demographic Parity, Disparate Impact, EU AI Act, Explainable AI, Facial Analysis, Fairness Metrics, NIST AI RMF, Surrogate Explainability / SHAP Fallback

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
BCa	Bias-Corrected and Accelerated
CLI	Command-Line Interface
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CSV	Comma-Separated Values
DIR	Disparate Impact Ratio
DPD	Demographic Parity Difference
EOD	Equalized Odds Difference
EOP	Equal Opportunity Difference
EU	European Union
F1	F1 Score
FR	Functional Requirement
GPU	Graphics Processing Unit
HTML	HyperText Markup Language
ITA	Individual Typology Angle
JSON	JavaScript Object Notation
NFR	Non-Functional Requirement
NIST	National Institute of Standards and Technology
PDF	Portable Document Format
RAM	Random-Access Memory
RMF	Risk Management Framework
SHAP	SHapley Additive exPlanations (deferred in favor of surrogate attribution)
SOLID	Single Responsibility, Open-Closed, Liskov Substitution, Interface Segregation, Dependency Inversion
UI	User Interface

VRAM	Video Random-Access Memory

LIST OF SYMBOLS

α	Significance threshold for chi-squared testing of independence between the predicted-label distribution and the demographic subgroup ($\alpha = 0.05$, NFR-001)
n	Subgroup sample size; the minimum threshold below which a metric is flagged as statistically insufficient rather than reported ($n \geq 30$, NFR-003)

1. INTRODUCTION

1.1. Background

Automated facial analysis systems, covering perceived gender classification, age estimation, and race or ethnicity prediction, are now embedded across commercial and public-sector applications ranging from photo tagging and demographic analytics to surveillance and access control. As these systems have proliferated, a substantial body of empirical research has documented that many of them exhibit statistically significant accuracy disparities across demographic subgroups, particularly by skin tone, gender presentation, and age. The foundational audit by Buolamwini and Gebru demonstrated that leading commercial gender-classification services performed markedly worse on darker-skinned female subjects than on lighter-skinned male subjects, with error rates as high as 34.7% for the worst-performing intersectional subgroup against as low as 0.8% for the best-performing one [1]. That finding catalysed a wider research agenda into fairness auditing for computer vision, subsequently surveyed comprehensively by Dehdashtian et al., who catalogue the sources of bias that arise across the machine learning pipeline, from data collection and annotation through model training and deployment, and the corresponding families of fairness metrics and mitigation techniques proposed in response [2].

Despite this growing awareness, independent, systematic, and reproducible auditing of facial analysis models remains comparatively rare in practice. Most organisations that deploy such systems either do not audit them for demographic bias at all, or rely on ad hoc, non-standardised evaluations that are difficult to reproduce, compare, or scrutinise externally. Dehdashtian et al.'s survey confirms that while a rich taxonomy of fairness metrics and mitigation techniques exists in the computer vision literature, comparatively little attention has been paid to standardising the auditing workflow itself into a repeatable pipeline [2].

This gap is now closing a regulatory window rather than an academic one. The European Union (EU) Artificial Intelligence (AI) Act (EU Regulation 2024/1689) enters into force for Article 10 (Data and Data Governance) on 2 August 2026, imposing binding data-governance, representativeness, and bias-detection obligations on providers of high-risk AI systems, a category that explicitly includes biometric categorisation [3]. The National Institute of Standards and Technology (NIST) AI Risk Management Framework (Risk Management Framework (RMF)) offers a complementary, voluntary structure, organised around four iterative functions, Govern, Map, Measure, and Manage, for operationalising exactly this kind of risk assessment [4]. Organisations that have not embedded documentation and measurement infrastructure into their development lifecycle ahead of this deadline face substantially higher retrofit costs than those that plan for it from the outset.

1.2. Problem Statement

There is therefore a clear engineering gap between a mature and growing body of fairness metrics, benchmark datasets, and bias-mitigation research, and the availability of an accessible, reusable, standards-aligned software platform that operationalises this research into a repeatable auditing workflow for facial analysis systems specifically. Existing audits, including the Gender Shades study itself, are typically conducted externally, manually, and for a single point in time; they are not designed as reusable, repeatable software artefacts that other researchers or organisations can apply to arbitrary models. This leaves model owners, auditors, and researchers without a systematic way to evaluate a facial-analysis model's performance across demographic subgroups using standardised benchmark datasets, statistically grounded fairness metrics, and a transparent, shareable report, at precisely the moment regulatory frameworks such as the EU AI Act begin to require exactly this kind of evidence.

1.3. Objectives

1.3.1. General Objective

To design and propose an end-to-end diagnostic and evaluative software platform, BiasAperture, that systematically identifies demographic accuracy disparities in third-party facial analysis models and reports them in a standardised, regulator-legible format.

1.3.2. Specific Objectives

1. To design a modular software architecture that accepts a trained facial analysis model, or a file of its precomputed predictions, together with a demographically labelled benchmark dataset, and computes subgroup and intersectional fairness metrics from it.
2. To integrate established open-source fairness toolkits, AIF360 and Fairlearn, alongside standard statistical significance testing, so that every disparity the platform reports is methodologically defensible rather than an artefact of sampling noise.
3. To generate a standardised, exportable fairness report, structurally informed by the Model Cards [5] and Datasheets for Datasets [6] documentation conventions, that communicates subgroup performance disparities in a form accessible to both technical and non-technical stakeholders.

4. To validate the platform's design against publicly available, demographically annotated benchmark datasets; the current case study uses FairFace [7], while UTKFace was profiled and cut from the implementation scope.
5. To map every computed fairness metric to a specific obligation under Article 10 and Annex IV of the EU AI Act and to the corresponding function of the NIST AI RMF, so the platform's output is directly usable as compliance evidence.

1.4. Scope and Limitations

1.4.1. Scope

BiasAperture is scoped as a strictly diagnostic and evaluative platform. In scope are: (i) demographic subgroup and intersectional performance evaluation of a supplied facial-analysis classifier; (ii) computation of four disparity metrics, Demographic Parity Difference (DPD), Equalized Odds Difference (EOD), Equal Opportunity Difference (EOP), and Disparate Impact Ratio (DIR), backed by chi-squared significance testing and bootstrap confidence intervals; (iii) explainability to help identify which visual features are associated with a detected disparity — the current explainability implementation uses demographic-dummy surrogate attribution, and richer spatial SHapley Additive exPlanations (deferred in favor of surrogate attribution) (SHAP) and ITA analysis remain deferred; and (iv) automated generation of a Model-Cards- and Datasheets-style HyperText Markup Language (HTML) fairness report, with each metric traceable to its EU AI Act and NIST AI RMF basis. The current case study uses FairFace [7]; UTKFace was profiled and cut from the implementation scope, with a FairFace-trained Convolutional Neural Network (CNN) baseline serving as the minimum case study.

1.4.2. Limitations

Consistent with these boundaries, BiasAperture does not attempt to mitigate bias, retrain or fine-tune models, or generate synthetic demographic data; its function is strictly diagnostic, not corrective. It reports where disparities exist and their statistical strength, but it does not prescribe or apply any remediation. Its findings are only as reliable as the label quality of the benchmark dataset supplied; the current case study uses FairFace, since UTKFace, whose model-estimated age labels were a known limitation, was profiled and cut from the implementation scope, and every reported subgroup below a minimum sample size of $n \geq 30$ is flagged as statistically insufficient rather than scored. Full-dataset evaluation is bounded by a documented runtime target rather than guaranteed instantaneous; and if project timeline pressure requires descoping, the platform's own cut-list (Chapter 4) defines, in order, which secondary capabilities are dropped first, while the diagnostic core, ingestion, one model interface, the fairness engine, one report format, and the scope-boundary statement itself, is never cut.

2. LITERATURE REVIEW

Chapter 1 established that while algorithmic fairness has evolved into a mature academic discipline, computer vision practitioners still lack a standardized, reusable, and statistically grounded auditing framework. To bridge this divide, this chapter conducts an in-depth synthesis of the scientific, mathematical, and regulatory literature that underpins the architecture of BiasAperture.

Rather than treating the literature as an uncurated catalogue of citations, we organize this review across seven coherent thematic pillars:

1. **The Empirical Mandate**: Foundational evidence demonstrating persistent demographic and intersectional disparities in facial analysis systems;
2. **Mathematical Formulations of Fairness**: Formal definitions of error-rate and selection-rate parities, along with critiques of borrowed legal thresholds;
3. **Statistical Rigor and Demographic Inference Epistemology**: Uncertainty quantification, sample-size guards, and the mathematical bounds imposed by demographic inference accuracy;
4. **Explainability Foundations and Attribution Bounds**: Cooperative game theory, additive feature attribution, and impossibility theorems bounding post-hoc explanations;
5. **Standardized Documentation and Regulatory Governance**: Recognized reporting conventions and legal-to-technical verification mappings;
6. **The Pipeline and Workflow Void**: The structural gap between theoretical metrics and reusable, air-gapped auditing software; and
7. **Frontiers and Future Diagnostic Extensions**: Emerging directions in dataset-level bias profiling, contextual-intersectional stress testing, and continuous attribute partitioning.

Table 2-1 at the conclusion of this chapter consolidates these twenty contributing works, highlighting their specific architectural tier and operational role within BiasAperture.

2.1. The Empirical Mandate for Facial Analysis Bias Auditing

The empirical foundation of demographic bias auditing in computer vision was established by Buolamwini and Gebru in their seminal *Gender Shades* study [1]. Auditing commercial gender-classification APIs across skin-type and gender intersections, the authors demonstrated that high aggregate accuracies masked severe subgroup disparities: while lighter-skinned males experienced error rates as low as 0.8%, darker-skinned females suffered error rates as high as 34.7%. This work proved that aggregate performance metrics are fundamentally insufficient for high-stakes biometric systems, establishing intersectional stratification as a mandatory evaluation paradigm. BiasAperture directly operationalizes this insight by automating intersectional cohort extraction across multi-demographic composite keys ($7 \times 2 \times 9$ strata), transforming what was originally an ad-hoc, manual empirical inquiry into a standardized software pipeline.

While *Gender Shades* audited model outputs against explicit demographic groups, Kurian et al. investigated the internal mechanisms through which convolutional networks acquire bias [8]. Utilizing controlled synthetic medical imaging, the authors demonstrated that deep networks learn to exploit demographic characteristics through unrelated visual features—such as image texture, brightness, and background lighting—even when demographic labels are entirely absent during training. This finding reveals that detected subgroup disparities cannot simply be presumed to stem from direct label correlations; they may instead arise from proxy variable entanglement. BiasAperture addresses this vulnerability by incorporating a targeted explainability layer: whenever a demographic disparity is statistically flagged, the platform initiates surrogate attribution to evaluate whether proxy features drive the disparity before reporting it as evidence.

Recent work by Nemavhola et al. reinforces this empirical mandate across modern computer vision backbones [9]. Evaluating ResNet-50, MobileNetV3, and Vision Transformer (DeiT) architectures on FairFace and UTKFace, the authors demonstrated that racial disparities consistently dominate gender disparities by an order of magnitude, producing large effect sizes (Cohen's $d > 1.0$) across all evaluated models. Crucially, their findings proved that newer architectures and higher overall accuracies do not alleviate subgroup disparity gaps. This provides direct empirical justification for BiasAperture's architectural premise: fairness auditing is an essential, independent validation step that cannot be substituted by general model scaling or architectural refinement.

2.2. Mathematical Formulations of Group Fairness and Disparity

To evaluate algorithmic fairness objectively, the machine learning community has formulated mathematical criteria that formalize what it means for a system to be fair. Hardt et al. introduced two foundational error-rate parity criteria in supervised learning: *Equalized Odds* and *Equal Opportunity* [10]. Equalized Odds mandates that a classifier produce equal True Positive Rates (TPR) and equal False Positive Rates (FPR) across all demographic cohorts, conditioning predictions on the true ground-truth label:

$$P(\hat{Y} = 1 \mid A = a, Y = y) = P(\hat{Y} = 1 \mid A = b, Y = y), \quad \forall y \in \{0, 1\}, \forall a, b. \qquad (2\text{-}1)$$

Equal Opportunity relaxes this criterion to require parity in TPR alone for the advantageous outcome ($Y = 1$). Hardt et al. proved that these criteria answer distinct ethical and statistical questions, such that neither subsumes the other. BiasAperture implements both definitions directly as two pillars of its **Core Four** disparity metrics: Equalized Odds Difference (EOD) and Equal Opportunity Difference (EOP), harmonizing their multi-group max-of-gaps definitions across independent computing backends.

However, naive adoption of fairness thresholds presents severe epistemic hazards. Watkins et al. delivered a critical analysis of the U.S. Equal Employment Opportunity Commission's "four-fifths rule" (the 80% disparate impact ratio), describing its uncritical importation into computer science as an instance of "epistemic trespassing" [11]. The authors demonstrated that treating an arbitrary ratio threshold (0.80) as a binary pass/fail test grants a false sense of compliance while obfuscating underlying sample size dynamics and variance. BiasAperture directly adopts Watkins et al.'s critique as an architectural invariant: the platform never evaluates Disparate Impact Ratio (DIR) or Demographic Parity Difference (DPD) against a bare, uncontextualized numerical threshold. Instead, every reported ratio and difference is paired with asymptotic significance testing and non-parametric bootstrap confidence intervals.

Further scrutinizing metric stability, Howard et al. conducted a comprehensive meta-evaluation of face-recognition fairness measures against Functional Fairness Measure Criteria (FFMC), focusing on intuitive net contribution, bounded ranges, and calculability at zero-error edge cases [12]. Their analysis revealed that popular metrics such as the False Distribution Ratio (FDR) and Inequity Rate (IR) suffer from mathematical fragility, severe range compression, and uncalculability when error rates approach zero. Howard et al. demonstrated that algorithmic fairness rankings flip dramatically depending on metric choice. This provides rigorous justification for BiasAperture's dual-backend cross-validation strategy: by executing **Fairlearn** and **AIF360** in parallel, reconciling their zero-denominator edge-case contracts ($0/0 \rightarrow 1.0$, $x/0 \rightarrow 0.0$), and raising explicit divergence warnings ($|\Delta| > 0.01$), BiasAperture protects auditors against single-metric artifacts. While Howard et al.'s metric framing is tailored to biometric verification error rates (FMR/FNMR), BiasAperture adapts these stability principles to classification disparity metrics.

2.3. Statistical Rigor, Uncertainty, and Demographic Inference Epistemology

A persistent vulnerability in algorithmic auditing is the reliance on unverified point estimates. Nemavhola et al. demonstrated that empirical face-attribute evaluations frequently report subgroup disparities without uncertainty quantification, leading practitioners to overstate apparent fairness or overreact to sample noise [9]. To establish statistical reliability, their work paired disparity measurements with bootstrap confidence intervals, analysis of variance (ANOVA), and Cohen's d effect sizes. BiasAperture builds upon this methodology by incorporating a vectorized, stratified Bias-Corrected and Accelerated (Bias-Corrected and Accelerated (BCa)) bootstrap engine ($B \geq 1,000$ resamples), calculating 95% confidence intervals for every Core Four metric alongside Pearson's χ^2 test of independence (with Fisher's exact test fallback for sparse 2×2 contingency tables).

Crucially, statistical reliability depends on adequate sample support. When subgroup sample sizes fall below critical thresholds, variance explodes and disparity metrics become wildly distorted. BiasAperture enforces a strict invariant ($n < 30$): any demographic subgroup with fewer than 30 observations is automatically suppressed from disparity computation, returning `metric_value = None` and flagging `insufficient_sample = True`.

Beyond sample variance, demographic auditing faces a deeper epistemic challenge: the provenance and reliability of demographic labels themselves. In a groundbreaking study, Fournier-Montgieux et al. proved that fairness-audit validity is mathematically bounded by the error rate of the Demographic Attribute Inference (DAI) process used to label protected groups [13]. Formulating formal bias and variance bounds, the authors proved that errors in demographic labeling directly corrupt downstream fairness metrics. Most significantly, Fournier-Montgieux et al. documented that FairFace's published training pipeline exhibits an open, unresolved reproducibility gap and that its ResNet-34 classifier trails improved demographic inference heads by 7 to 10 accuracy points.

This finding establishes a critical epistemic boundary for BiasAperture. In the locked Milestone M1 architecture, BiasAperture utilizes Karkkainen and Joo's FairFace dataset [7]—comprising 97,698 released images across a balanced 7-race, 2-gender, and 9-age taxonomy—as its primary benchmark, while excluding UTKFace due to synthetic DEX model-estimated age noise and collapsed racial categories. However, pursuant to the findings of Fournier-Montgieux et al., BiasAperture's audit reports explicitly treat FairFace labels as an audited operational benchmark rather than infallible ground truth, documenting that demographic inference errors represent an unavoidable upper bound on audit certainty.

2.4. Explainability Foundations and Bounds on Feature Attribution

When a facial analysis system exhibits significant demographic disparity, auditors must determine whether the model is relying on justifiable visual features or impermissible proxies. The theoretical foundation of feature attribution originates in cooperative game theory. Lloyd Shapley established that for an n-player cooperative game, there exists exactly one payoff allocation that simultaneously satisfies four fundamental axioms: *Efficiency* (full distribution of the surplus), *Symmetry* (equal rewards for equal marginal contributions), *Dummy/Null Player* (zero reward for non-contributing players), and *Additivity* (linear separability across games) [14]:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} \left[v(S \cup \{i\}) - v(S) \right] . \tag{2-2}$$

Lundberg and Lee unified this game-theoretic formulation with additive feature attribution, proposing the SHAP (SHapley Additive exPlanations) framework and deriving the optimal Shapley kernel for weighted linear regression [15]:

$$\pi_x(z') = \frac{M - 1}{\binom{M}{|z'|} \cdot |z'| \cdot (M - |z'|)} . \tag{2-3}$$

In BiasAperture, exact Shapley values ground our surrogate explainability implementation. Rather than attempting computationally intractable pixel-level Shapley estimation across high-resolution facial rasters ($50,176$ features per 224×224 image), BiasAperture implements an additive demographic surrogate attribution model ($X \to \hat{Y}_{\text{pred}}$ via linear logistic regression). Using the exact additive relationship $\phi_i = w_i \cdot (x_i - \mathbb{E}[x_i])$, the engine attributes model predictions across protected and proxy demographic axes whenever a disparity is statistically flagged ($p < 0.05, n \geq 30$). Full spatial feature attribution (via `shap.PartitionExplainer`) and Individual Typology Angle (Individual Typology Angle (ITA)) colorimetry remain documented deferred capabilities.

Critically, BiasAperture approaches feature attribution with strict epistemic caution, informed by two seminal theoretical critiques. Bilodeau et al. proved a series of impossibility theorems demonstrating that no feature attribution method satisfying completeness and linearity can reliably separate true causal features from spurious correlations in general deep neural networks [16]. Simultaneously, Slack et al. proved that post-hoc explainers like LIME and SHAP can be circumvented by adversarial model scaffolding that detects perturbation distributions and disguises discriminatory logic during audit runs [17]. In accordance with Bilodeau et al. and Slack et al., BiasAperture establishes a mandatory reporting policy: surrogate attribution findings are reported strictly as identifying observable proxy correlations, never as certifying the absence of bias or proving causal invariance.

2.5. Standardized Documentation, Governance, and Regulatory Compliance

A diagnostic audit is ineffective if its findings cannot be communicated clearly to multidisciplinary stakeholders and regulatory authorities. BiasAperture grounds its report generation engine in established documentation conventions and emerging regulatory standards.

Mitchell et al. proposed *Model Cards for Model Reporting*, establishing a standardized documentation format detailing a model's intended domain, performance characteristics, evaluation datasets, and ethical considerations [5]. Complementing this, Gebru et al. formulated *Datasheets for Datasets*, mandating structured disclosure of dataset provenance, collection mechanisms, annotation protocols, and known demographic limitations [6]. BiasAperture's zero-network HTML reporting engine adopts these twin conventions directly, structuring audit outputs into dedicated model performance profiles and dataset provenance sections (documenting the FairFace benchmark and formally recording the rationale for excluding UTKFace).

On the regulatory frontier, Buscemi et al. formulated a systematic methodology for decomposing the high-risk requirements of the European Union Artificial Intelligence Act (Regulation EU 2024/1689 [3], as amended by the Digital Omnibus Regulation EU 2026/1744 [18]) into concrete, technology-agnostic technical verification activities [19]. BiasAperture adopts Buscemi et al.'s decomposition methodology to implement a clause-level compliance crosswalk (Section 4.5). Specifically, the platform maps each Core Four metric, statistical test, and sample-size guard directly to Article 10(2)(f) (bias examination and mitigation), Article 10(3) (dataset representation), and Article 13 (transparency and interpretability).

Furthermore, BiasAperture aligns its diagnostic controls with the *Measure* core function of the National Institute of Standards and Technology AI Risk Management Framework (NIST AI RMF 1.0) [4]. Audit metrics directly instrument Measure 2.11 (evaluating fairness and bias across demographic cohorts), Measure 1.1 (formal documentation of sample-size limitations and unmeasurable cohorts), and Measure 1.3 (rigorous validation via independent dual backends).

2.6. The Pipeline and Workflow Void in Computer Vision Auditing

In their comprehensive survey of fairness and bias mitigation in computer vision, Dehdashtian et al. mapped the state of the discipline across the machine learning lifecycle: data collection, curation, annotation, model training, and post-deployment monitoring [2]. Their analysis uncovered a profound structural asymmetry: while the academic literature provides an extensive catalogue of theoretical fairness metrics and algorithmic mitigation techniques (such as adversarial de-biasing, contrastive loss re-weighting, and fair representation learning), there is a striking absence of standardized, reusable software pipelines designed specifically to execute multi-group computer vision audits repeatably. Existing fairness toolkits (such as Fairlearn and AIF360) were developed primarily for tabular datasets, requiring extensive custom wrapping and manual data engineering to ingest computer vision pipelines.

BiasAperture is architected specifically to eliminate this workflow void. Rather than introducing another bespoke mitigation technique or competing mathematical metric, BiasAperture operates strictly as an independent diagnostic and evaluative platform. It provides a turn-key CLI orchestration engine that automates dataset ingestion, enforces schema invariants, executes cross-validating dual-backend fairness computation, calculates rigorous statistical significance and bootstrap confidence intervals, initiates targeted surrogate explainability, and compiles standalone, air-gapped HTML compliance reports. By deliberately declining to perform model retraining, weights debiasing, or synthetic augmentation, BiasAperture preserves absolute diagnostic neutrality.

2.7. Frontiers and Future Diagnostic Extensions

To maintain rigorous diagnostic relevance over extended development cycles, BiasAperture's research foundation incorporates three advanced works that delineate the future frontiers of algorithmic bias auditing.

First, Cascone et al. established a formal framework for evaluating representational and stereotypical bias at the dataset level, prior to and independent of model training [20]. Auditing CelebA, LFW, and MAAD-Face, the authors demonstrated that representational disparities collapse to a single latent factor, whereas stereotypical attribute correlations vary widely across datasets. This work motivates a future **Tier 2** extension for BiasAperture: an upstream dataset-profiling module that audits training and validation corpora for representational skew and stereotypical correlation before model predictions are ingested.

Second, Aslam et al. introduced *CIFA* (Contextual-Intersectional Fairness Auditing), demonstrating that traditional demographic-only audits fail to uncover severe "hidden subgroup" performance degradations caused by environmental and contextual factors [21]. Evaluating models across combined demographic and contextual axes (such as illumination variations, motion blur, and facial accessories), CIFA revealed performance drops of over 26% in contextual subgroups that aggregate audits completely masked. While CIFA's proposed mitigation re-training is out of scope for BiasAperture's diagnostic mandate, its contextual-intersectional evaluation paradigm represents a direct candidate for future diagnostic stress testing.

Finally, Shilova et al. addressed the limitations of fixed categorical demographic schemas in their *FairGroups* framework [22]. The authors demonstrated that quantizing continuous biological attributes—such as skin tone represented in continuous CIELAB color space—into coarse, predefined discrete bins (such as the 6-point Fitzpatrick scale) can obscure substantial intra-group disparities. Using data-driven partitioning, FairGroups identified vulnerable subgroups that standard discrete bins missed. For BiasAperture, whose Milestone M1 schema is locked to discrete categorical axes ($7 \times 2 \times 9$), Shilova et al.'s work provides the theoretical blueprint for a future continuous-attribute auditing module that evaluates continuous skin reflectance (ITA) alongside discrete demographic taxonomies.

2.8. Summary of Reviewed Literature

Table 2-1 consolidates the twenty works analyzed in this chapter, detailing their core contributions, their architectural tier, and their exact operational adoption within the BiasAperture diagnostic platform.

Table 2-1. Comprehensive literature review matrix: 20 contributing works mapped to BiasAperture architectural tiers, core contributions, and adopted concepts.

Authors	**Paper Title (Year)**	**Key Contribution**	**Architectural Tier & Adopted Concepts**
Theme 1: Empirical Foundations of Intersectional Bias			
Buolamwini & Gebru [1]	Gender Shades (2018)	Audits commercial facial analysis APIs; reveals error rates up to 34.7% on darker females vs 0.8% on lighter males.	**Core Foundation (Tier 1).** Motivated automated intersectional cohort stratification ($7 \times 2 \times 9$) over aggregate accuracy metrics.
Kurian et al. [8]	Bias Encoding in CNNs (2024)	Proves CNNs encode demographic information as proxy variables via neutral visual features (texture, lighting).	**Core Foundation (Tier 1).** Direct justification for targeted explainability: surrogate attribution evaluates proxy reliance on flagged disparities.

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Authors	Paper Title (Year)	Key Contribution	Architectural Tier & Adopted Concepts
Nemavhola et al. [9]	Reassessing Demographic Bias (2026)	Multi-model audit on FairFace/UTKFace; proves racial disparities dominate gender gaps ($d > 1.0$) across modern backbones.	**Core Foundation (Tier 1).** Validates that model scaling does not eliminate bias; establishes empirical priors for FairFace evaluation.
Theme 2: Mathematical Formulations of Fairness			
Hardt et al. [10]	Equality of Opportunity (2016)	Formulates Equalized Odds (TPR+FPR parity) and Equal Opportunity (TPR parity) in supervised learning.	**Core Foundation (Tier 1).** Directly adopted as the mathematical definitions of EOD and EOP in the Core Four metric battery.
Watkins et al. [11]	Four-Fifths Rule Critique (2022)	Critiques importing employment law thresholds (80%) into ML as epistemically unjustified bare ratios.	**Core Foundation (Tier 1).** Governs NFR-001: DIR is never reported as a bare ratio; always paired with χ^2 testing and bootstrap CIs.
Howard et al. [12]	Evaluating Proposed Fairness Models (2022)	Evaluates FR fairness metrics against FFMC criteria; demonstrates metric instability and ranking inversions.	**Supporting Literature (Tier 1).** Grounds dual-backend cross-validation (Fairlearn + AIF360) and divergence monitoring ($\lvert\Delta\rvert > 0.01$).
Theme 3: Statistical Rigor & Demographic Epistemology			
Nemavhola et al. [9]	Statistical Grounding in Bias (2026)	Demonstrates that point-estimate fairness audits overstate certainty; mandates bootstrap CIs and effect sizes.	**Core Foundation (Tier 1).** Grounds BiasAperture's vectorized BCa bootstrap engine ($B \geq 1{,}000$) and $n \geq 30$ sample guards.
Fournier-			

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Authors	**Paper Title (Year)**	**Key Contribution**	**Architectural Tier & Adopted Concepts**
Montgieux et al. [13]	Reliable Demographic Inference (2025)	Proves fairness audit validity is bounded by demographic inference quality; cites FairFace reproducibility gaps.	**Core Validity Threat (Tier 1).** Codified as Claim R-021: FairFace labels are audited as benchmark references, acknowledging DAI error bounds.
Karkkainen & Joo [7]	FairFace Benchmark (2021)	Constructs balanced 97,698-image face dataset with 7-race, 2-gender, and 9-age categories to minimize skew.	**Core Benchmark (Tier 1).** Primary validation dataset (FR-001); UTKFace excluded due to synthetic age noise and race collapsing.
Theme 4: Explainability Foundations & Theoretical Bounds			
Shapley [14]	Value for n-Person Games (1953)	Axiomatic cooperative game theory proving Shapley value is the unique credit allocation satisfying four fairness axioms.	**XAI Foundation (Tier 1).** Theoretical bedrock of additive feature attribution framed as fair surplus allocation over baseline expectations.
Lundberg & Lee [15]	Unified Approach to Interpretation (2017)	Unifies additive feature attribution (SHAP); derives optimal Shapley kernel for weighted linear surrogate regression.	**XAI Foundation (Tier 1).** Methodological foundation for demographic surrogate explainer and deferred spatial SHAP (`Partition-Explainer`).
Bilodeau et al. [16]	Impossibility Theorems (2022)	Proves no linear/complete attribution method can reliably separate true causal drivers from spurious correlations.	**XAI Theoretical Bound (Tier 1).** Governs reporting language: surrogate findings indicate proxy correlations, not causal proof of non-bias.

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Authors	Paper Title (Year)	Key Contribution	Architectural Tier & Adopted Concepts
Slack et al. [17]	Fooling LIME and SHAP (2020)	Demonstrates adversarial scaffolding can disguise discriminatory model behavior from post-hoc explainers.	**XAI Threat Model (Tier 1).** Bounds trust in post-hoc XAI; motivates dual model interfacing (predictions file vs in-process inference).
Theme 5: Standardized Documentation & Governance			
Mitchell et al. [5]	Model Cards (2019)	Proposes standardized, stakeholder-legible documentation format for model performance and intended use.	**Core Reporting (Tier 1).** Adopted as structural foundation for the model performance evaluation section of generated HTML reports.
Gebru et al. [6]	Datasheets for Datasets (2018)	Proposes structured documentation for dataset provenance, collection mechanisms, and demographic composition.	**Core Reporting (Tier 1).** Adopted as structural foundation for the dataset provenance section; documents benchmark inclusion/cut criteria.
Buscemi et al. [19]	Assessing High-Risk AI Systems (2025)	Decomposes EU AI Act high-risk obligations into concrete, technology-agnostic technical verification activities.	**Regulatory Crosswalk (Tier 1).** Methodological template for Article 10(2)(f), 10(3), and 13 technical compliance mapping (Section 4.5).
NIST [4]	AI Risk Management Framework (2023)	Consensus risk management taxonomy organizing AI governance into Govern, Map, Measure, and Manage functions.	**Regulatory Crosswalk (Tier 1).** Direct instrumentation of Measure 2.11 (bias auditing), Measure 1.1 (limitations), and Measure 1.3.
Theme 6: Pipeline Engineering & Diagnostic Tooling			

Continued on next page

Authors	Paper Title (Year)	Key Contribution	Architectural Tier & Adopted Concepts
Dehdashtian et al. [2]	Fairness in CV: A Survey (2024)	Surveys bias sources across CV pipeline; reveals rich mitigation literature but immature reusable auditing pipelines.	**Workflow Gap (Tier 1).** Identifies the exact engineering void BiasAperture closes: a reusable, diagnostic-only, air-gapped pipeline.
Theme 7: Frontiers & Future Diagnostic Extensions			
Cascone et al. [20]	Bias-Aware Dataset Evaluation (2026)	Taxonomy distinguishing model-output bias from dataset-level representational and stereotypical attribute bias.	**Future Extension (Tier 2).** Motivates candidate upstream dataset-profiling module prior to and distinct from model inference audits.
Aslam et al. [21]	CIFA: Contextual Auditing (2026)	Demonstrates demographic audits mask severe errors under contextual variations (illumination, blur, accessories).	**Future Extension (Tier 2).** Informs future contextual-intersectional stress testing layer; mitigation steps excluded per diagnostic scope.
Shilova et al. [22]	FairGroups: Continuous Variables (2025)	Data-driven partitioning of continuous sensitive attributes (skin tone in CIELAB) to expose hidden intra-group bias.	**Future Extension (Tier 2).** Theoretical basis for future continuous skin-tone (ITA) evaluation layer on top of locked discrete M1 schema.

By synthesizing these twenty foundational, empirical, statistical, explainability, and governance works, BiasAperture establishes an integrated diagnostic framework that bridges the gap between academic fairness research and actionable, regulator-legible compliance engineering. Chapters 3 and 4 formulate these principles into formal system requirements and architectural specifications.

3. REQUIREMENT ANALYSIS

This chapter specifies what BiasAperture must do, its Functional Requirements (FRs), and how well it must do it, its Non-Functional Requirements (NFRs). Hardware and software requirements are given as *system requirements*: what platform and environment are needed to build and run the system. Chapter 4 explains *how* these requirements are implemented and validated.

3.1. Functional Requirements

Functional requirements are listed using IEEE 830-style "shall" statements, one capability per requirement.

1. **FR-001 (Data Ingestion):** The system shall load, validate, and standardise a benchmark dataset (principally FairFace; UTKFace was profiled and cut from the implementation scope) or a compatible custom dataset, checking image integrity and aligning demographic annotation fields (race or ethnicity, perceived gender, age group) into a common internal schema.
2. **FR-002 (Dual-Mode Model Interface):** The system shall obtain predictions from a target facial-analysis model either by direct in-process inference against a supplied PyTorch or TensorFlow model object, or by batch ingestion of a precomputed predictions file in Comma-Separated Values (CSV) or JavaScript Object Notation (JSON) format.
3. **FR-003 (Fairness Metric Computation):** The system shall compute, per subgroup and per demographic intersection, the four Core Four disparity metrics, DPD, EOD, EOP, and DIR, using AIF360 and Fairlearn as independent, cross-validating computational backends.
4. **FR-004 (Statistical Significance Testing):** The system shall accompany every reported disparity with a chi-squared test of independence between the predicted-label distribution and the demographic subgroup, and with a bootstrap confidence interval on the subgroup accuracy difference.
5. **FR-005 (Explainability):** The system shall generate surrogate feature-attribution output (with spatial SHAP analysis deferred) identifying which input features are most associated with a detected disparity, to support diagnosis of proxy-variable entanglement.
6. **FR-006 (Report Generation):** The system shall assemble the computed metrics, statistical tests, and dataset and model provenance information into a standardised, exportable HTML fairness report, structurally informed by the Model Cards and Datasheets for Datasets conventions, with every metric row showing subgroup

sample size, point estimate, 95% confidence interval, and either a p-value or an insufficient-sample flag.

7. **FR-007 (Regulatory Traceability):** The system shall map every computed metric, in the generated report, to its corresponding clause under Article 10 or Annex IV of the EU AI Act and to its corresponding function within the NIST AI RMF.
8. **FR-008 (Orchestration and Configuration):** The system shall expose a command-line configuration interface through which an auditor selects the benchmark dataset, the model or predictions source, and the metrics to compute, and shall re-run identically given the same configuration and inputs.
9. **FR-009 (Licensing Acknowledgement):** The system shall display and require explicit acknowledgement of the selected dataset's licensing terms before an audit is executed.

3.2. Non-Functional Requirements

1. **NFR-001 (Statistical Rigour):** All significance testing shall use a threshold of α = 0.05, and the system shall report exact p-values rather than only a pass/fail flag.
2. **NFR-002 (Uncertainty Quantification):** Every reported subgroup accuracy difference shall include a 95% confidence interval computed from a minimum of 1,000 bootstrap resamples.
3. **NFR-003 (Data-Integrity Guard):** Any subgroup with a sample size below n = 30 shall be automatically flagged as "insufficient sample, not reported" rather than assigned a computed metric value.
4. **NFR-004 (Performance):** A full-dataset evaluation over FairFace's 97,698 released images (86,744 train + 10,954 validation) shall complete within 4 hours on a single mid-range Graphics Processing Unit (GPU); a stratified development subset of n = 5,000 images shall complete within 30 minutes on Central Processing Unit (CPU) alone.
5. **NFR-005 (Modularity):** The system shall maintain strict separation of concerns across its architectural modules, each exposing a narrow, well-defined interface, so that any module can be unit-tested and extended independently of the others.
6. **NFR-006 (Reproducibility):** Bootstrap resampling shall use fixed random seeds, and third-party dependency versions (AIF360, Fairlearn, PyTorch or TensorFlow) shall be pinned in a lock-file to prevent silent behavioural drift between runs.
7. **NFR-007 (Portability):** The system shall run cross-platform on any environment supporting the pinned Python dependency stack, without requiring proprietary or platform-specific components.

8. **NFR-008 (Explainability Performance):** Surrogate feature attribution (with spatial SHAP analysis deferred) shall be computed only on flagged disparities rather than the full dataset by default, to keep explainability overhead a small fraction of total runtime.

3.3. System Requirements

This section answers what machine and software stack are required to build and run BiasAperture.

3.3.1. Hardware Requirements

- **Minimum (development, stratified subset):** 64-bit CPU, 4 or more cores, 8 GB Random-Access Memory (RAM), 5 GB free disk space.
- **Recommended (full-dataset audit, NFR-004 target):** A single mid-range GPU with 8 GB or more Video Random-Access Memory (VRAM), for example an NVIDIA T4-class device, either a local workstation GPU or a free-tier cloud notebook GPU such as Google Colab.
- No specialised or custom hardware is required; the platform performs inference and metric computation only, never model training from scratch.

3.3.2. Software Requirements

- **Operating system:** Linux, Windows, or macOS; the stack is pure Python and cross-platform.
- **Language and runtime:** Python 3.10 or later.
- **Model interface:** PyTorch or TensorFlow, whichever framework the audited model was built in.
- **Image preprocessing:** OpenCV, pandas, NumPy.
- **Fairness computation:** AIF360, Fairlearn, scikit-learn, SciPy.
- **Explainability:** Demographic-dummy surrogate attribution (scikit-learn; SHAP deferred).
- **Report generation:** Jinja2 for HTML templating.

4. SYSTEM ARCHITECTURE AND METHODOLOGY

Chapter 3 specified *what* BiasAperture must do. This chapter specifies *how*: the architectural decomposition and design principles that satisfy those requirements, the mapping from computed metrics to specific regulatory clauses, the work breakdown and schedule under which the system is to be built, and the verification strategy that gives the resulting fairness findings statistical credibility.

4.1. Design Principles

The architecture follows Object-Oriented Software Engineering practice under Single Responsibility, Open-Closed, Liskov Substitution, Interface Segregation, Dependency Inversion (SOLID) guidelines, applied concretely rather than generically, so that each principle maps onto a specific structural decision rather than remaining an abstract design ideal. The single-responsibility principle is realised directly by the five-module split in Section 4.2: each module owns exactly one concern, ingestion, model access, metric computation, explanation, or reporting, and no module reaches into another's internal state. The open-closed principle is realised by the Strategy pattern around the fairness-metrics back end (Section 4.4): the Core Four disparity metrics, and the AIF360 and Fairlearn backends that compute them, are implemented as interchangeable strategies behind a common `FairnessBackend` interface, so a future fifth disparity metric, or a replacement fairness library, is added by implementing a new strategy without modifying the computation engine's internals. The Liskov substitution and interface segregation principles are satisfied by the narrowness of these same interfaces: because `ModelInterface` and `FairnessBackend` each expose only the minimal operations their callers use, any concrete implementation, whether `InProcessInterface`, `PredictionsFileInterface`, `AIF360Backend`, or `FairlearnBackend`, can be substituted for another without altering the correctness of the code that depends on the interface. The dependency-inversion principle is realised by having the engine and report generator depend only on the abstract `FairnessBackend` and `ModelInterface` interfaces, never on AIF360, Fairlearn, or a specific file format directly, so that a cut or substitution at either boundary (Section 4.8) changes an implementation, not a contract. Finally, the shared schema described in Section 4.7 is an application of interface-first, design-by-contract practice: the field set two independently developed streams must both honour is fixed before either stream begins substantive work, allowing the two streams to be built and unit-tested in parallel against a shared specification rather than against each other's evolving implementation.

4.2. Architectural Modules

This section presents the system architecture of BiasAperture and the specific modules that realise it. The platform is composed of a set of cooperating modules, each responsible for one stage of the audit pipeline, coordinated by a central orchestration layer. Fig. 4-1 depicts this architecture as a whole, showing how external datasets and target models enter the system, how they are processed through the pipeline, and how the resulting compliance report is produced.

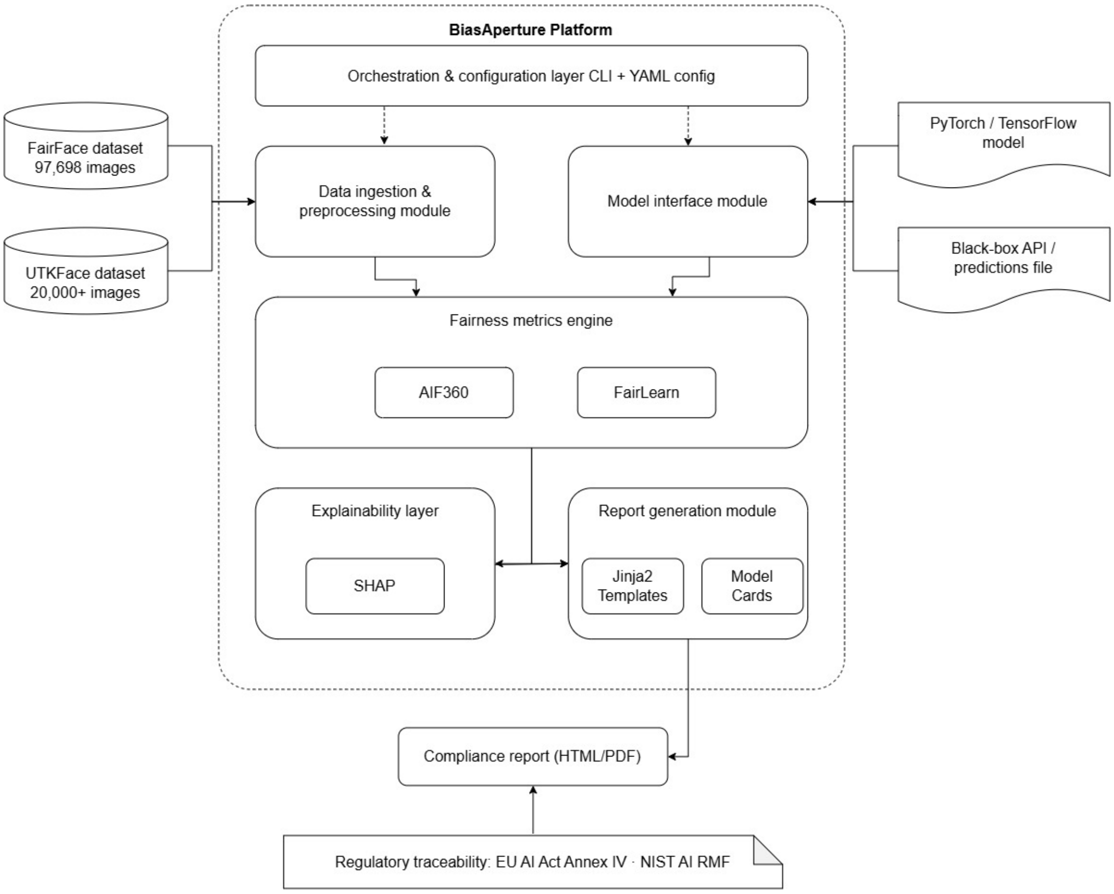

Figure 4-1. BiasAperture system architecture.

Table 4-1 formalises the architecture shown in Fig. 4-1, listing each module alongside its primary responsibility, the functional requirements it satisfies, and the key libraries it depends on. The remainder of this section discusses each module in turn, following the order in which data and control flow through the pipeline.

Table 4-1. Architectural modules and their responsibilities.

Module	Primary Responsibility	Key Libraries
Data Ingestion and Preprocessing	Load, validate, and standardise a benchmark or custom dataset (FR-001)	OpenCV, pandas, NumPy
Model Interface	Obtain predictions via direct inference or a predictions file (FR-002)	PyTorch, TensorFlow
Fairness Metrics Engine	Compute subgroup and intersectional metrics with significance testing (FR-003, FR-004)	AIF360, Fairlearn, scikit-learn, SciPy
Explainability Layer	Attribute a detected disparity to input features (FR-005)	SHAP
Report Generation	Assemble findings into a Model-Cards- and Datasheets-structured report (FR-006, FR-007)	Jinja2
Orchestration and Configuration	Coordinate the pipeline end to end behind a single entry point (FR-008, FR-009)	Python Command-Line Interface (CLI)

4.2.1. Data Ingestion and Preprocessing Module

This module loads a selected benchmark dataset — with the current case study using FairFace, since UTKFace was profiled and cut from the implementation scope — or a compatible custom dataset, validates image integrity, standardises image resolution and colour space via OpenCV, and aligns demographic annotation fields (race or ethnicity, perceived gender, age group) into the locked internal schema (Section 4.7). It rejects corrupted or unreadable files and enforces that every image carries a complete, correctly typed set of demographic labels before any downstream computation proceeds.

4.2.2. Model Interface Module

This module is a thin, framework-agnostic adapter that obtains predictions from a target model without requiring internal weight access. Two integration modes are supported: direct in-process inference against a supplied PyTorch or TensorFlow model exposing a standard predict interface, and batch ingestion of a precomputed predictions file in CSV or JSON format for models that cannot be executed locally. The module performs no training, fine-tuning, or weight modification of any kind.

4.2.3. Fairness Metrics Computation Engine

The analytical core of the platform. It computes per-subgroup and intersectional performance metrics (accuracy, precision, recall, F1 Score (F1), false-positive and false-negative rates) and the four Core Four disparity metrics, DPD, EOD, EOP, and DIR, using AIF360 and Fairlearn as independent, cross-validating computational back ends: because both toolkits already implement, test, and document these metrics, the platform does not need to re-derive fairness statistics from first principles, and any discrepancy between the two backends' output on the same input surfaces an integration defect rather than a genuine result. The engine additionally performs the chi-squared significance testing and bootstrap resampling required by FR-004.

4.2.4. Explainability Layer

Motivated by the proxy-variable finding discussed in Chapter 2, this layer applies SHAP to input features associated with a disparity the metrics engine has flagged, producing feature-attribution output that helps distinguish a genuinely demographic effect from an artefact of an unrelated correlated variable such as lighting or background. Consistent with NFR-008, attribution is computed only on flagged disparities rather than across the full dataset by default.

4.2.5. Report Generation Module

This module assembles the computed metrics, statistical tests, and dataset and model provenance into a standardised, exportable HTML report, structurally informed by the Model Cards and Datasheets for Datasets conventions discussed in Chapter 2. Every metric row shows subgroup sample size, point estimate, 95% confidence interval, and either a p-value or an insufficient-sample flag (FR-006), and every row is additionally traceable to its Article 10 or Annex IV clause and its NIST AI RMF function (FR-007, Section 4.5). Because the reporting layer consumes only the already-computed metrics dictionary, it is architecturally decoupled from the computation engine, which simplifies testing and prevents a reporting defect from affecting the correctness of the underlying analysis.

4.2.6. Orchestration and Configuration Layer

A lightweight layer coordinates the pipeline end to end, dataset selection, model connection, metric selection, and report generation, behind a single entry point, and exposes a configuration file and command-line interface through which an auditor specifies the audit parameters (FR-008). The dataset's licensing terms must be displayed and explicitly acknowledged before an audit is executed (FR-009).

4.3. End-to-End Workflow of the System

Fig. 4-2 presents the end-to-end workflow followed by the system during a single audit run, tracing the sequence of steps and decision points from initial configuration through to the generation of the final compliance report.

The workflow begins with the auditor configuring the audit: specifying the benchmark or custom dataset, the target model, or a predictions file, and the metrics to compute. The system then branches on the model access mode. If a model object is supplied, predictions are obtained by direct in-process inference against the PyTorch or TensorFlow model; otherwise, predictions are obtained by batch ingestion of a precomputed CSV or JSON file. Both branches converge on a common path, in which the selected dataset — the current case study uses FairFace, since UTKFace was profiled and cut from the implementation scope — is ingested and validated, and predictions are generated over the resulting test matrix.

With predictions and ground-truth labels in hand, the system computes the Core Four disparity metrics together with demographic-dummy surrogate feature attribution (with richer spatial SHAP deferred). Before any metric is reported, each demographic subgroup is checked against the minimum sample-size guardrail: subgroups with fewer than thirty observations are flagged as an insufficient sample and excluded from the statistical

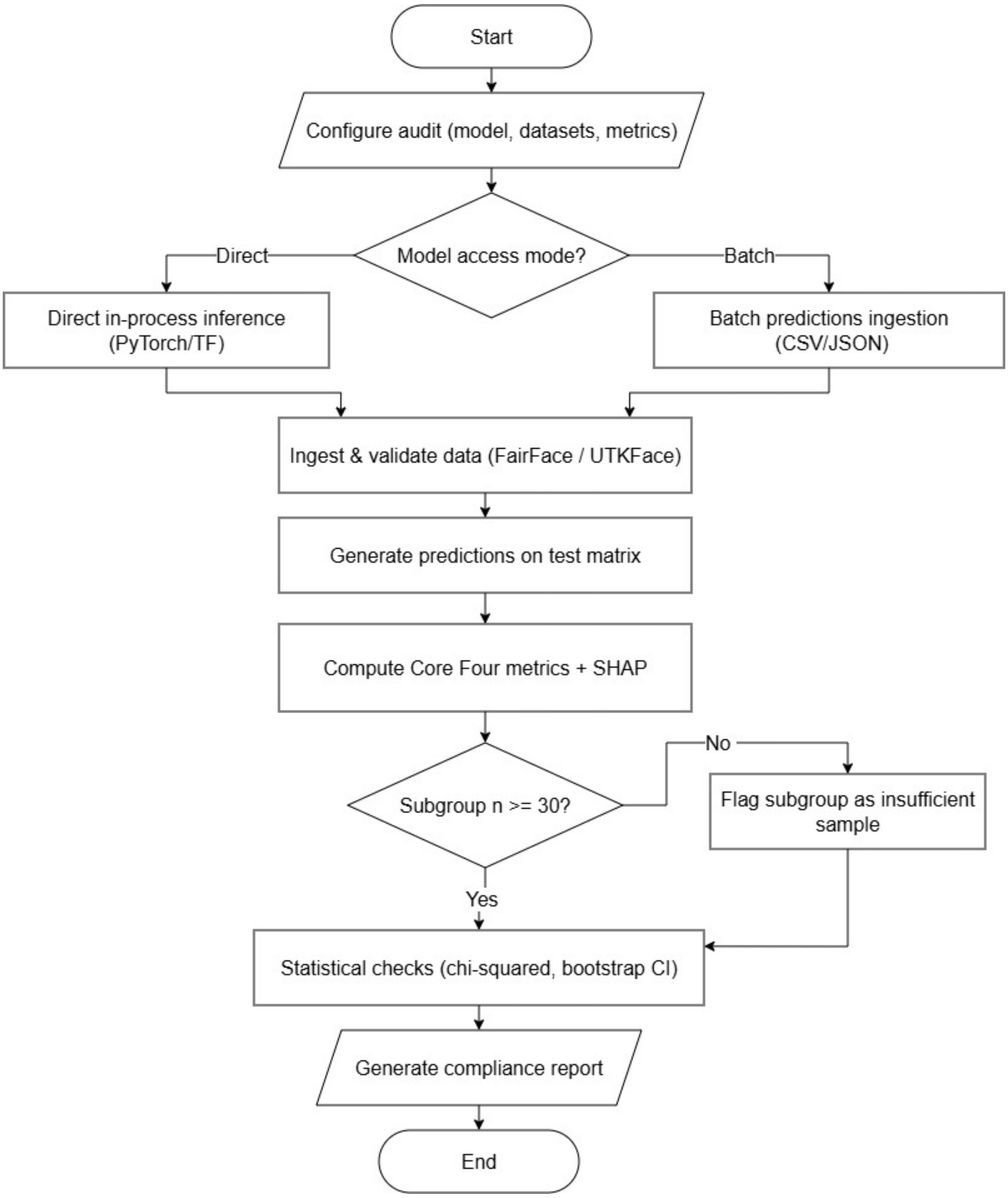

Figure 4-2. BiasAperture end-to-end audit workflow.

checks that follow, while all other subgroups proceed directly. The remaining subgroups are then subjected to chi-squared significance testing and bootstrap confidence-interval estimation, after which the system assembles and generates the final compliance report, concluding the workflow.

4.4. Design Patterns

Each pattern in Table 4-2 is tied to a specific project mechanic, either a descoping decision (Section 4.8) or the shared-schema contract (Section 4.7), rather than applied for its own sake.

Table 4-2. Design patterns mapped to architectural modules.

Pattern	Module	Structure	Rationale
Adapter	Model Interface	`ModelInterface` → `InProcessInterface` / `PredictionsFileInterface`	Cutting direct inference (Section 4.8, item 4) means simply not instantiating one interface; downstream code is unchanged
Strategy	Fairness Metrics Engine	`FairnessBackend` → `AIF360Backend` / `FairlearnBackend`	Cutting AIF360 (Section 4.8, item 5) is a backend swap, not a rewrite; the four metrics are interchangeable strategies
Builder	Data Ingestion	`DataIngestionPipeline`	Matches the multi-step curation (acquisition, integrity check, schema alignment) the module performs
Factory	Report Generation	`HTMLReportGenerator`	Keeps the HTML report path untouched regardless of which export formats are in scope
Facade	Orchestration	`CrossValidationOrchestrator`	Hides the ingestion → interface → engine → report sequence behind one call

4.5. Regulatory Traceability Mapping

FR-007 requires that every computed metric be mapped to a specific regulatory basis. Table 4-3 shows how the EU AI Act's Article 10 data-governance components map to the platform's own outputs, following the operational-verification approach of Buscemi et al. discussed in Chapter 2 [3, 19].

The NIST AI RMF's four functions map onto the project less as a report artefact than as a description of what the platform, and the process that built it, together accomplish. *Govern* is satisfied by the scope-boundary statement (Section 4.8) and TA-reviewed

Table 4-3. Article 10 components mapped to BiasAperture outputs.

Article 10 Component	Technical Requirement	BiasAperture Evidence
10(2) Governance	Documentation of design choices and data origin	Dataset provenance and licensing-acknowledgement log (FR-009)
10(3) Quality	Representativeness and error minimisation	Subgroup sample-size reporting and the $n \geq 30$ insufficiency guard (NFR-003)
10(4) Context	Geographical and behavioural setting adjustment	Per-dataset provenance section of the exported report
10(5) Mitigation	Bias detection and disparity evidence	The four Core Four disparity metrics with significance testing (FR-003, FR-004)

feasibility study that establish accountability for what the platform will and will not claim to do. *Map* is satisfied by the explicit demographic schema (Section 4.7) that identifies which subgroups the platform is designed to evaluate. *Measure* is the function the platform most directly operationalises: the fairness engine applies the Core Four metrics with significance testing to determine whether a subgroup disparity meets the defined statistical thresholds. *Manage* is out of the platform's diagnostic scope by design, since BiasAperture reports disparities rather than prioritising or remediating them, and this exclusion is itself stated explicitly in every generated report.

4.6. Work Breakdown Structure

Table 4-4 decomposes the project into deliverable-oriented work packages following the 100% rule, every module in Section 4.2 is covered and nothing is added outside stated scope, with leaf packages sized for a two-person team and each leaf tied either to an acceptance criterion in Chapter 3 or to a module in Section 4.2. Ownership is tagged *Together* where a package requires shared judgement or iteration, or *Stream A* / *Stream B* where it can proceed solo; named assignment of which team member takes which solo stream remains open at time of writing and will be settled by demonstrated preference rather than fixed in advance.

4.7. Project Plan and Schedule

The work packages in Table 4-4 are restructured for concurrency rather than executed as a strict sequence, so that neither team member is idle while the other's phase is blocking. Figure 4-3 shows the resulting eight-week schedule.

Table 4-4. Work breakdown structure.

ID	Work Package	Ownership
1.1	Project management: feasibility review, acceptance-criteria lock, schema lock, submission and defence prep	Together
1.2	Classifier selection and model interface: baseline confirmation, `ModelInterface` abstraction, predictions-file ingestion path	Together
1.3	Stream A, test-matrix construction: dataset acquisition and integrity check, inference run, schema curation, stratified development subset	Stream A
1.4	Stream B, report scaffolding: HTML and Jinja2 template, Model Cards / Datasheets structure, hand-written mock metrics dictionary	Stream B
1.5	Statistical detection engine: backend integration, four disparity metrics, chi-squared testing, bootstrap confidence interval, sample-size guard	Together
1.6	Integration: Stream A output into the detection engine, mock-to-real swap of Stream B's report against real findings, orchestration module	Together
1.7	Verification and validation: runtime targets, report-completeness rule, case-study run	Together
1.8	Documentation and delivery: scope-boundary statement, final report, submission package	Together

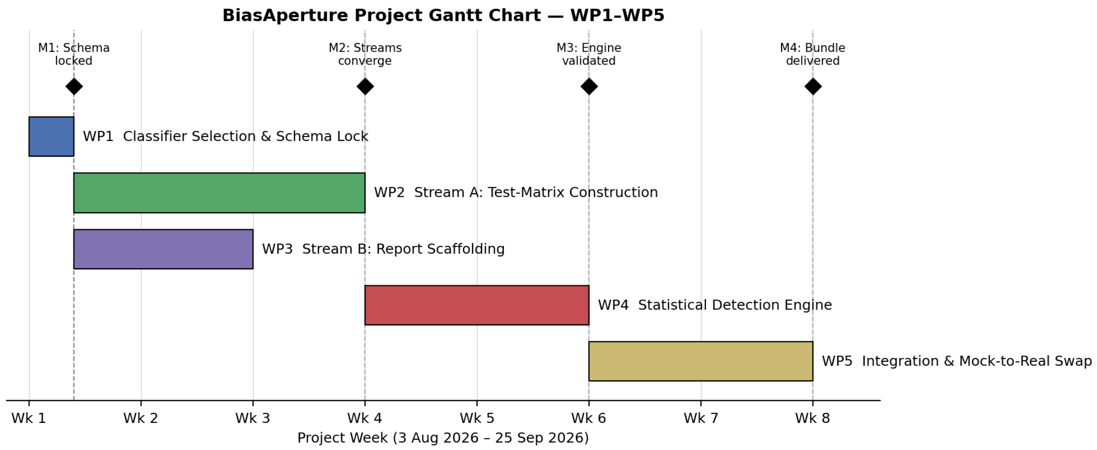

Figure 4-3. BiasAperture project schedule, work packages WP1–WP5.

WP1, *Classifier Selection and Schema Lock*, is a short, foundational, jointly worked phase that fixes the classifier baseline and the internal demographic schema every later module must honour: an `image_id`, race, gender, and age subgroup fields, predicted and true labels, and, for the detection engine's output, metric name, point estimate, confidence bounds, p-value, and subgroup sample size. Its completion is Milestone M1, *schema locked*.

With the schema fixed, WP2 (Stream A, test-matrix construction) and WP3 (Stream B, report scaffolding) begin immediately and run in parallel rather than sequentially: Stream A curates the FairFace-based dataset and predictions file against the locked schema while Stream B builds the HTML report template and Model Cards / Datasheets structure against a hand-written mock metrics dictionary that validates against the same field set. Both streams converge at Milestone M2.

WP4, the *Statistical Detection Engine*, is the joint convergence phase: it consumes Stream A's schema-aligned test matrix as input and produces the schema-shaped metrics dictionary Stream B's report template already expects, running faster than it otherwise would because both interfaces were fixed before this phase began. Its completion, Milestone M3, is the point at which the engine is validated end to end.

WP5, *Integration and Mock-to-Real Swap*, replaces Stream B's mock metrics dictionary with the detection engine's real output, a short, mechanical step precisely because the template was built against an identical schema in week one, and completes with the final case-study audit and report bundle at Milestone M4.

4.8. Descoping Strategy

The scope defined in Chapter 3 is deliberately fuller than a two-person, eight-week timeline is guaranteed to accommodate without slippage. Rather than discover a descoping order under deadline pressure, Table 4-5 fixes it in advance.

Table 4-5. Descoping order if the project falls behind schedule.

#	Cut	Rationale
1	Web User Interface (UI), keep CLI only	Pure user-experience layer; no bearing on the fairness-engineering core
2	Cut UTKFace, keep FairFace only	The architecture already anticipates this cut; UTKFace's DEX-estimated label noise is an already-documented risk
3	Portable Document Format (PDF) export, keep HTML only	HTML alone satisfies the standardised, exportable report requirement and the Model Cards / Datasheets structure
4	Direct in-process inference, keep predictions-file ingestion only	The simpler mode; removes GPU and dependency-version risk entirely, and a public pretrained checkpoint and inference script make it sufficient for the case study
5	AIF360, keep Fairlearn only	Last resort; Fairlearn alone computes all four Core Four metrics, but this is the only cut that measurably weakens the cross-validated, methodologically defensible claim

The non-negotiable core that is never cut, regardless of timeline pressure, is: data ingestion, one model interface, the fairness engine, one report format, and the scope-boundary statement itself.

4.9. Verification and Validation Strategy

Statistical correctness is verified before any real-data run: the chi-squared and bootstrap-resampling routines are unit-tested against known reference values first, so that a defect in the significance-testing code is caught independently of whatever pattern happens to appear in a real audit's results. Both streams' outputs are additionally schema-validated against the locked field set from Section 4.7, directly defusing the risk that independent development on either side silently drifts and breaks the WP5 integration step. Version control follows a feature-branch-per-stream model off a `main` branch holding only the locked schema and module skeletons, with review at the WP4 convergence point before merge. Runtime and report-completeness are validated against the acceptance criteria already fixed in Chapter 3 (NFR-002, NFR-003, NFR-004), and the case study itself, a FairFace-trained ResNet-34 baseline evaluated on all four Core Four metrics, exercises the full pipeline end to end as the platform's own validation instance.

5. CONCLUSION

5.1. Summary

This report has proposed BiasAperture, a diagnostic and evaluative software platform for auditing demographic accuracy disparities in third-party facial analysis models, and specified it in enough detail to be built against by a two-person team over an eight-week schedule. Chapter 2 showed that a mature fairness-metrics and mitigation-techniques literature exists without a corresponding reusable auditing artefact, at precisely the moment the EU AI Act's Article 10 obligations make that gap a compliance question rather than only an academic one. Chapters 3 and 4 then specified how the platform closes it, against each of Chapter 1's five specific objectives in turn: a modular architecture built around FR-001 and FR-002 satisfies the first; AIF360 and Fairlearn as independent, cross-validating backends for the four Core Four metrics, with chi-squared and bootstrap significance testing, satisfy the second; a report structured against the Model Cards and Datasheets conventions surveyed in Chapter 2 satisfies the third; validation against the current case study, which uses FairFace since UTKFace was profiled and cut from the implementation scope, with a FairFace-trained CNN baseline as the minimum case study, satisfies the fourth; and the Article 10 and NIST AI RMF mapping detailed in Section 4.5 satisfies the fifth.

Consistent with the scope stated in Chapter 1, BiasAperture remains strictly diagnostic throughout this design: it reports where a disparity exists and how statistically strong it is, and it does not mitigate bias, retrain a model, or generate synthetic demographic data. Section 4.8's descoping order exists precisely so that this diagnostic core, ingestion, one model interface, the fairness engine, one report format, and the scope-boundary statement itself, is the one part of the proposal that is never at risk under timeline pressure, whatever else is cut first.

5.2. Future Work

The descoping order in Table 4-5 doubles as a map of the platform's nearest extension points once the eight-week schedule in Section 4.7 is complete: a web UI over the existing CLI, PDF export alongside the HTML report, direct in-process inference restored alongside predictions-file ingestion, and UTKFace (currently cut per Cut-List #2) brought back in as a fully supported secondary benchmark once its label-noise limitation, discussed in Chapter 2, is more thoroughly characterised. Beyond the current cut-list, the clearest longer-term direction follows from the diagnostic and mitigation split Dehdashtian et al.'s survey draws [2]: BiasAperture is deliberately positioned on the diagnostic side of that split, and a natural next stage of the same research programme is a mitigation-facing companion that consumes BiasAperture's findings as input rather than folding mitigation into the diagnostic platform itself, keeping the separation of concerns this proposal establishes intact rather than eroding it.

A. PROJECT BUDGET

BiasAperture is a pure-software diagnostic platform with no bill of materials or physical hardware deliverable (Chapter 3, System Requirements); the budget below is scoped accordingly to cloud compute, subscriptions, and documentation costs rather than component procurement.

A.1. Cloud Computing and Development Services

Table A-1. Cloud computing and development services cost.

Service	Description	Cost
Open-source software	Python, PyTorch/TensorFlow, AIF360, Fairlearn, scikit-learn, SciPy, OpenCV, SHAP (surrogate fallback), Jinja2	Free
Cloud GPU compute (NFR-004 target)	Google Colab, free-tier T4-class GPU for full-dataset evaluation runs	Free
Cloud GPU compute (contingency)	Google Colab Pro, if free-tier quota is exhausted during the full 97,698-image FairFace evaluation (NFR-004)	$10/month
Cloud storage	Google Drive, dataset and checkpoint storage during development	Free (existing quota)
	Subtotal	**$0–$10/month**

A.2. Documentation and Miscellaneous

Table A-2. Documentation and miscellaneous cost.

Category	Item / Description	Estimated Cost
Printing and binding	Proposal, progress, and final report copies for submission and defence	$10–$15
Miscellaneous	Contingency for unplanned documentation or minor tooling needs	$5–$10
	Total Estimated Cost	**$15–$35 one-time, plus $0–$10/month cloud compute contingency**

B. PROJECT TIMELINE

The project is executed over the eight-week schedule specified in Section 4.7, structured into work packages WP1–WP5 as detailed in Table 4-4. Figure B-1 reproduces the Gantt chart of Fig. 4-3 for appendix reference, showing task durations, the Stream A / Stream B concurrency, and milestones M1–M4.

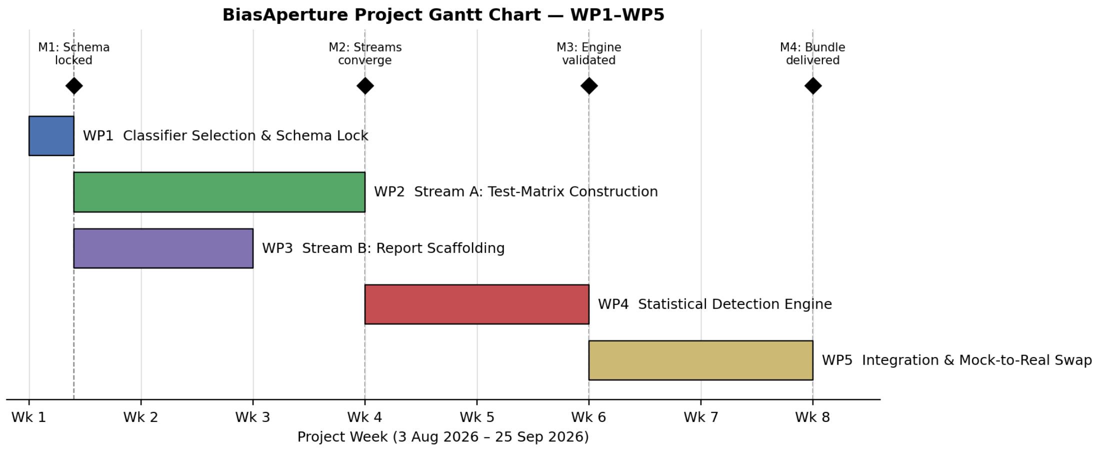

Figure B-1. BiasAperture project schedule, work packages WP1–WP5 (reproduced from Chapter 4).

C. LOCKED INTERNAL SCHEMA

Section 4.7 fixes the internal schema every module must honour before Stream A and Stream B begin parallel work. Tables C-1 and C-2 give the exact field-level specification of that schema.

Table C-1. Dataset / test-matrix schema.

Field	Type	Description
`image_id`	string	Unique identifier for the source image
`race`	categorical (7 levels)	FairFace race / ethnicity category
`gender`	categorical (binary)	Perceived gender label
`age`	categorical (9 bins)	FairFace age-group bin
`true_label`	categorical	Ground-truth demographic attribute value
`predicted_label`	categorical	Value returned by the Model Interface module
`protected_attribute`	categorical	Subgroup field currently selected for disparity analysis

Table C-2. Metrics-dictionary schema, the shape every Fairness Metrics Computation Engine output must satisfy.

Field	Type	Description
`metric_name`	string	One of DPD, EOD, EOP, DIR
`subgroup_n`	integer	Subgroup sample size
`point_estimate`	float	Computed metric value
`ci_lower`, `ci_upper`	float pair	95% bootstrap confidence interval bounds ($\geq$1,000 resamples)
`p_value`	float	Result of the chi-squared significance test
`insufficient_sample_flag`	boolean	True when subgroup $n < 30$ (NFR-003)

D. RISK REGISTER

Table D-1 reproduces the risk register from the feasibility study underlying this proposal, covering the risks judged material enough to shape a design decision elsewhere in this report.

Table D-1. Project risk register: likelihood, impact, and mitigation strategy.

Risk	Likelihood	Impact	Mitigation
Demographic label noise, particularly UTKFace's model-estimated age labels	Medium	Medium	Formally cut UTKFace per Cut-List #2; report subgroup n and 95% confidence intervals alongside every metric; cross-validate against FairFace's manually annotated labels; document the limitation explicitly
Model interface incompatibility with non-standard prediction formats	Medium	Medium	Provide the documented CSV / JSON predictions-file ingestion path as a fallback to direct in-process inference
Misinterpretation of metrics by non-specialist stakeholders	Medium	High	Plain-language explanation for every metric, with explicit caveats on statistical significance and sample size, in the generated report
Scope creep toward mitigation or model retraining	Medium	High	Written scope document reviewed with the supervisor; report generation restricted to diagnostic, non-prescriptive output only
Licensing non-compliance for a custom user-supplied dataset	Low	High	Display and require explicit acknowledgement of licensing terms before an audit executes (FR-009)
Computational resource constraints delaying full-dataset evaluation	Low	Medium	Stratified development subset (n = 5,000) supported for development; expected hardware runtimes documented (NFR-004)
Dependency version drift across AIF360, Fairlearn, and the model framework	Medium	Low	Dependency versions pinned in a lock-file; periodic re-validation against upstream releases (NFR-006)
Dual-use risk: findings used to exploit rather than remediate a model's weaknesses	Low	Medium	Intended, permitted use stated explicitly in project documentation; platform restricted to diagnostic reporting only

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